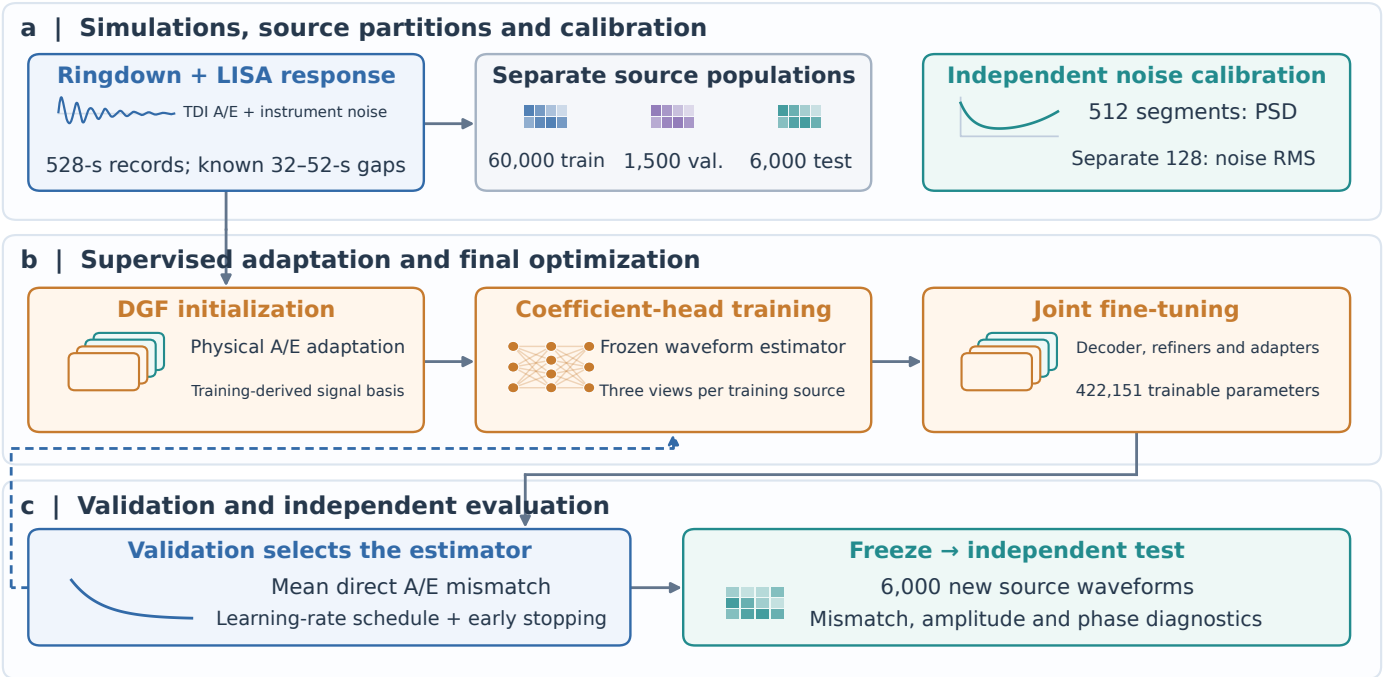
a | Simulations, source partitions and calibration
Ringdown + LISA response
TDI A/E + instrument noise
528-s records; known 32–52-s gaps
Separate source populations
60,000 train
1,500 val.
6,000 test
Independent noise calibration
512 segments: PSD
Separate 128: noise RMS
b | Supervised adaptation and final optimization
DGF initialization
Physical A/E adaptation
Training-derived signal basis
Coefficient-head training
Frozen waveform estimator
Three views per training source
Joint fine-tuning
Decoder, refiners and adapters
422,151 trainable parameters
c | Validation and independent evaluation
Validation selects the estimator
Mean direct A/E mismatch
Learning-rate schedule + early stopping
Freeze → independent test
6,000 new source waveforms
Mismatch, amplitude and phase diagnostics